Q1 What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines and software to perform tasks that normally require human intelligence. These tasks include learning from data, recognizing images and speech, understanding language, solving problems, and making decisions.

Examples of AI

- Voice assistants such as Siri and Google Assistant
- Chatbots that answer questions
- Self-driving vehicle technology
- Face recognition systems
- Recommendation systems used by streaming and shopping platforms

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How AI Works

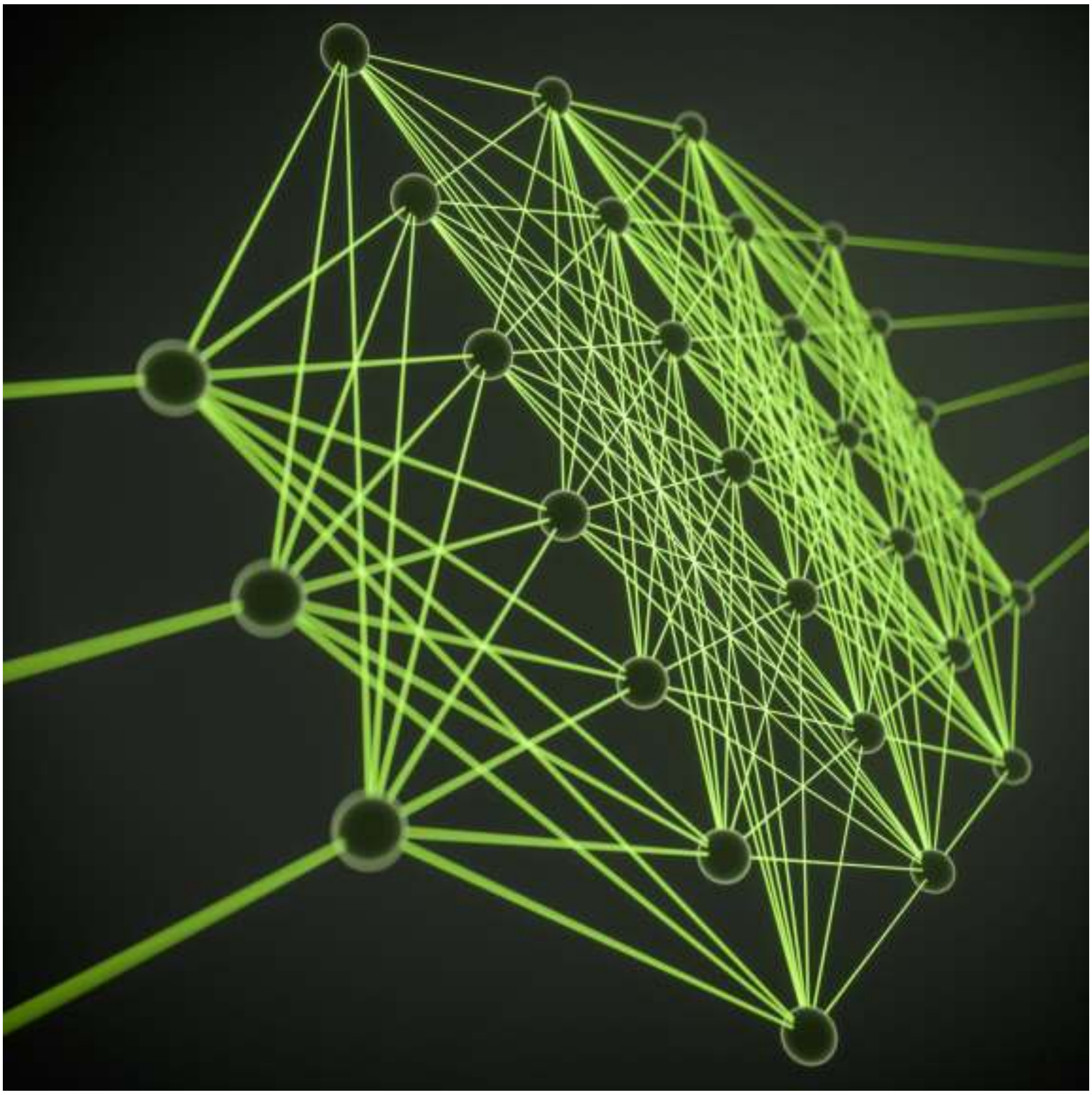
1. Collects data
2. Learns patterns from the data
3. Makes predictions or decisions
4. Improves performance with more experience and data

Q2 Difference between AI, Machine Learning, Generative AI, and Agentic AI.

Understanding the relationship between AI, Machine Learning, Generative AI, and Agentic AI is easiest if you think of them as nested categories:

erm	What It Is	Main Goal	Example
AI (Artificial Intelligence)	The broad field of creating machines that can perform tasks requiring human intelligence.	Make machines intelligent.	A chess-playing program.
Machine Learning (ML)	A subset of AI where systems learn patterns from data instead of	Learn from data and improve performance.	Spam email detection.

erm	What It Is	Main Goal	Example
	being explicitly programmed.		
Generative AI	A subset of ML that creates new content such as text, images, audio, or code.	Generate new content.	ChatGPT writing an essay or an AI creating artwork.
Agentic AI	AI systems that can plan, reason, take actions, and pursue goals with limited human supervision.	Achieve objectives autonomously.	An AI assistant that books meetings, gathers information, and sends updates automatically.



Relationship

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Artificial Intelligence (AI)
│
├── Machine Learning (ML)
│   │
│   ├── Generative AI
│   │
```

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│    └── Other ML Applications
│
└── Agentic AI
```

Simple Examples

- **AI:** A calculator that solves problems intelligently.
- **Machine Learning:** A system that learns to recognize cats from thousands of cat photos.
- **Generative AI:** A system that creates a new cat image that never existed before.
- **Agentic AI:** A system that receives the goal "find the best cat food, compare prices, and order it" and carries out the steps on its own.

Q3 Explain any three open-source AI models available on Hugging Face.

1. Llama 3

Developer: Meta

What it does:

- **Understands and generates human-like text.**
- **Can answer questions, summarize documents, write code, and assist with conversations.**

Key features:

- **Strong reasoning and language understanding.**
- **Available in different sizes (e.g., 8B and 70B parameters).**
- **Can be fine-tuned for specific tasks.**

Applications:

- **Chatbots**
- **Virtual assistants**
- **Content generation**
- **Coding assistants**

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2. Mistral 7B

Developer: Mistral AI

What it does:

- **A compact yet powerful large language model designed for text generation and understanding.**

Key features:

- **Efficient and lightweight compared to larger models.**
- **Performs well on many language tasks.**
- **Lower computational requirements.**

Applications:

- **Customer support chatbots**
- **Text summarization**
- **Question answering**

- **Educational tools**

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3. Stable Diffusion-to-image generation model"]□

Developer：Stability AI

What it does:

- **Generates images from text descriptions.**
- **Converts written prompts into realistic or artistic images.**

Applications:

- **Digital art creation**
- **Graphic design**
- **Advertising content**

- ## Q4 What is the difference between traditional programming and AI-assisted development?

Aspect	Traditional Programming	AI-Assisted Development
How it works	Developers write all logic and code manually.	AI helps generate, explain, debug, and optimize code.
Development Speed	Usually slower because coding is done from scratch.	Faster due to code suggestions and automation.
Problem Solving	Relies entirely on developer expertise.	Combines developer expertise with AI-generated

Aspect	Traditional Programming	AI-Assisted Development
		recommendations.
Debugging	Developers find and fix errors manually.	AI can identify bugs and suggest fixes.
Learning Curve	Requires deeper knowledge of programming concepts from the start.	Beginners can get assistance while learning.
Code Quality	Depends solely on the developer's skills.	AI can suggest best practices, but human review is still needed.

Aspect	Traditional Programming	AI-Assisted Development
Creativity	Human-driven design and implementation.	AI assists with implementation, while humans guide goals and decisions.

Q5 Explain AI hallucinations with one example.

AI Hallucinations

An **AI hallucination** occurs when an AI system generates information that sounds believable and confident but is actually incorrect, fabricated, or unsupported by facts.

Why Do Hallucinations Happen?

- AI predicts likely words based on patterns in data.
- It does not inherently know whether a statement is true.
- If information is missing, ambiguous, or outside its knowledge, it may generate a plausible-sounding answer.

Example

User: "Who won the Nobel Prize in Physics in 2035?"

AI Hallucination:
"The 2035 Nobel Prize in Physics was awarded to Dr. Jane Smith for quantum teleportation research.

How to Reduce Hallucinations

- Verify important information using reliable sources.
- Ask AI to provide references or citations.
- Use up-to-date databases and fact-checking tools.
- Review AI-generated content before relying on it.